SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

FACULTY OF SCIENCE AND HUMANITIES

DEPARTMENT OF COMPUTER APPLICATIONS

PRACTICAL RECORD NOTE

STUDENT NAME :

REGISTER NUMBER :

CLASS : **Section :**

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

FACULTY OF SCIENCE AND HUMANITIES

DEPARTMENT OF COMPUTER APPLICATIONS

SRM Nagar, Kattankulathur – 603 203

CERTIFICATE

Certified to be the bonafide record of practical work done by

_____

Register No._____of_____Degree course for

UCA23D06J PROGRAMMING USING C# *in the Computer lab in SRM Institute of Science and Technology during the academic year 2025-2026.*

Staff In-charge *Head of the Department*

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EX NO: 01

DATE:

Program to demonstrate arithmetic operators

Aim :

Program to demonstrate arithmetic operators

Code:

```
using System; using
System.IO; using
System.Text;
namespace
IncludeHelp{
classTest{
static void Main(string[] args)  {
int a = 10;    int
b = 3;
int sum = a + b;
int sub = a - b;    int
mul = a * b;
float div = (float)a / (float)b;
int rem = a % b;
Console.WriteLine("Addition of {0} and {1} is = {2}", a, b, sum);
Console.WriteLine("Subtraction of {0} and {1} is = {2}", ,b, sub);
Console.WriteLine("Multiplication of {0} and {1} is = {2}", a, b, mul);
Console.WriteLine("Division of {0} and {1} is = {2}", a, b, div);
Console.WriteLine("Remainder of {0} and {1} is = {2}", a, b, rem);
Console.ReadLine();
        }
       }
      }
```

Output:

```
Addition of 10 and 3 is = 13
Subtraction of 10 and 3 is = 7
Multiplication of 10 and 3 is = 30
Division of 10 and 3 is = 3.333333
Remainder of 10 and 3 is = 1
```

Result:

The Program is executed successfully.

EX NO: 02
DATE:

Sum of Digits Program in C#

Aim :

To find Sum of Digits using C#

CODE:

```
namespace LogicalPrograms{
public class Program
{
    Static void Main(string[] args)
{
    Console.Write("Enter the Number : ");

 int number = int.Parse(Console.ReadLine());

int sum = 0, reminder;

while (number > 0)

        {

 reminder = number % 10;

sum = sum + reminder;

number = number / 10;}

Console.WriteLine("The Sum of Digits is : " + sum);

Console.ReadKey();

        }

                    }
```

OUTPUT:

```
Enter the Number : 738
The Sum of Digits is : 18
```

Result:

The program completed successfully.

EX NO: 03 **Program to print number pattern**

DATE:

Aim :

To write a program to print a number pattern

Program:

```
using System; class
Program {     static
          void
Main() {
      int  rows  =  5;
for (int i = 1; i
<= rows; i++) {
          for (int j = 1; j
<= i; j++) {

Console.Write(j);
          }

Console.WriteLine
();
       }
    } }
```

Output:

```
1
12
123
1234
12345
```

Result:

The pattern program is successfully completed.

EX NO: 04
DATE:

ARMSTRONG NUMBER

Aim:

To check whether a given three-digit integer is an Armstrong number.

Program:

```
using System;
class Program
{
static void Main()
{
int number = 153;
int temp = number;
int sum = 0;
while (number > 0) {
int r = number % 10;
sum = sum + r * r * r;
number = number / 10;
}
if (sum == temp)
Console.WriteLine("Armstrong Number");
else
Console.WriteLine("Not Armstrong Number");

}
}
```

Output:

Armstrong number

Result:

The Armstrong number check program is executed successfully.

EX NO: 05

DATE:

STRONG NUMBER

Aim :

To write a C# program to check if a number is a Strong number (Sum of factorials of digits).

Code:

```
Using System;
class Program {
static int factorial(int n) {
int f = 1;
for (int i = 1; i <= n; i++) f = f * i;
 return f;
}
static void Main() {
int number = 145;
int temp = number;
int sum = 0;
while (number > 0) {
int r = number % 10;
sum = sum + factorial(r);
number = number / 10; }
if(sum == temp)
   Console.WriteLine("Strong Number");
else
   Console.WriteLine("NotStrongNumber");
  }
  }
```

Output:

Strong Number

Result:

The Program is executed successfully.

EX NO: 06 **INHERITANCE PROGRAM**
DATE:

Aim :

To demonstrate the concept of Single Inheritance and Method Overriding in C#.

CODE:

```
using System;
class Shape {
    public virtual double Area() { return 0; }
}
class Triangle : Shape {
    double b, h;
    public Triangle(double baseLength, double height) {
        b = baseLength; h = height;
    }
    public override double Area() { return 0.5 * b * h; }
}
class Program {
    static void Main() {
        Triangle t = new Triangle(10, 5);
        Console.WriteLine("Triangle Area = " + t.Area());
    }
}
```

Output:

Triangle Area = 25

Result:

The program completed successfully.

EX NO: 07 **OPERATOR OVERLOADING**
DATE:

Aim :

To implement operator overloading for the '+' operator to add two objects.

Code:

```
using System;
class Number {
public int value;
public Number(int v) { value = v; }
public staticNumber operator +(Number a, Number b) {
return new Number(a.value + b.value);
}
}
class Program {
static void Main() {
Number n1 = new Number(10);
Number n2 = new Number(20);
Number result = n1 + n2;
Console.WriteLine("Result = " + result.value);
}
}
```

OUTPUT:

Result = 30

Result:

The operator overloading program executed successfully.

EX NO: 08
DATE:

DELEGATE PROGRAM

Aim:

To implement a delegate that points to addition and subtraction methods.

Program:

```
using System;
delegate int Operation(int a, int b);
class Program {
static int Add(int x, int y) { return x + y; }
static int Sub(int x, int y) { return x - y; }
static void Main() {
Operation op;
op = Add;
Console.WriteLine("Addition = " + op(10, 5));
op = Sub;
Console.WriteLine("Subtraction = " + op(10, 5));
}
}
```

Output:

```
Addition = 15
Subtraction = 5
```

Result:

The Delegate program is executed successfully.

EX NO: 09

DATE: **CUSTOM EXCEPTION**

Aim :

To create and throw a user-defined exception for insufficient bank balance.

Code:

```
using System;
class BalanceException :
Exception {
public BalanceException(string
message) : base(message) { }
}
class Program {
  static void Main() {
      double balance = 500;
      double withdraw = 700;
      try {
         if (withdraw > balance)
            throw new
BalanceException("Insufficient
balance");

balance=balancewithdraw;
      }
      catch (BalanceException
e) {
Console.WriteLine(e.Message);
}
   }
}
```

Output:

Insufficient Balance

Result:

The custom exception program is completed successfully

EX NO: 10 **RADIO BUTTON COLOR CHANGE**
DATE:

Aim :

To develop a Windows Form application that changes label color using RadioButtons.

CODE:

```
using System;
using System.Drawing;
using System.Windows.Forms;
public class Form1 : Form {
   Label label1 = new Label();
   RadioButton r1 = new RadioButton();
   RadioButton r2 = new RadioButton();
   RadioButton r3 = new RadioButton();
   public Form1() {
      label1.Text = "Change Color";
      label1.Location = new Point(50, 20);
      r1.Text = "Red"; r2.Text = "Green"; r3.Text = "Blue";
      r1.CheckedChanged += (s, e) => label1.ForeColor = Color.Red;
      r2.CheckedChanged += (s, e) => label1.ForeColor =Color.Green;
      r3.CheckedChanged += (s, e) => label1.ForeColor = Color.Blue;
      Controls.Add(label1); Controls.Add(r1);
      Controls.Add(r2); Controls.Add(r3);
   }
}
```

OUTPUT:

```
Enter the Number : 738
The Sum of Digits is : 18
```

Result:

The program completed successfully.

EX NO: 11 **program using RadioButton to**
Date: **change the bulb state ON/OFF.**

Aim:

To write a program using Radio Button to change the bulb state ON/OFF.

Code:

```
using System;
using System.Windows.Forms;

namespace WindowsFormsApp1
{
    public partial class Form1 : Form
  {
        public Form1()
 {
     InitializeComponent();
                    }

   private void radioButton1_CheckedChanged(object sender, EventArgs e)
{
        if (radioButton1.Checked)
  {
    pictureBox2.Show();
    pictureBox1.Hide();
                }
                }

    private void radioButton2_CheckedChanged(object sender, EventArgs e)
{
    if (radioButton2.Checked)
```

```
{
        pictureBox1.Show();
        pictureBox2.Hide();
       }
       }
       }
      }
```

Output:

Result:

The program using Radio Button to change the bulb state ON/OFF is completed successfully

EX NO: 12 **A Program to Change the Fore Color of**
DATE: **the text in Label**

Aim:

To Write a program to change the Fore Color of the text in Label using different RadioButtons such Red, Green, Blue.

Code:

```
using System;
using System.Drawing;
using System.Windows.Forms;

namespace WindowsFormsApp1
{
public partial class Form1 : Form
{
public Form1()
{
 InitializeComponent();
        }

 private void radioButton1_CheckedChanged(object sender, EventArgs e)
{
    if (radioButton1.Checked)
  {
  label1.ForeColor = Color.Red;
    }
  }

  private void radioButton2_CheckedChanged(object sender, EventArgs e)
  {
```

```
        if (radioButton2.Checked)
    {
      label1.ForeColor = Color.Green;
        }
    }

        private void radioButton3_CheckedChanged(object sender, EventArgs e)
   {
      if (radioButton3.Checked)
    {
      label1.ForeColor = Color.Blue;}}
          }}
```

Output:

Result:

A program to change the Fore Color of the text in Label using different RadioButtons such Red, Green, Blue is executed successfully.

EX NO: 13 **Develop Web Applications using**
DATE: **Data controls**

Aim:

To develop Web Applications using Data controls.

Code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Demo.aspx.cs" Inherits="Demo" %>

<!DOCTYPE html>
<html>
<head runat="server">
   <title>Demo Page</title>
</head>
<body>
   <form id="form1" runat="server">

      <h2>Demo Form</h2>

      <asp:Label ID="lblName" runat="server" Text="Name:"></asp:Label>
      <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
      <br /><br />

      <asp:Label ID="lblCity" runat="server" Text="City:"></asp:Label>
      <asp:ListBox ID="ListBox1" runat="server">
         <asp:ListItem>Mumbai</asp:ListItem>
         <asp:ListItem>Bangalore</asp:ListItem>
         <asp:ListItem>Hyderabad</asp:ListItem>
      </asp:ListBox>
      <br /><br />
```

```
        <asp:Label ID="lblGender" runat="server" Text="Gender:"></asp:Label>
        <asp:RadioButton ID="RadioButton1" runat="server" Text="Male" GroupName="gender" />
        <asp:RadioButton ID="RadioButton2" runat="server" Text="Female" GroupName="gender" />
        <br /><br />

        <asp:Label ID="lblSkills" runat="server" Text="Skills:"></asp:Label>
        <asp:CheckBox ID="CheckBox1" runat="server" Text="C#" />
        <asp:CheckBox ID="CheckBox2" runat="server" Text="ASP.NET" />
        <br /><br />

        <asp:Button ID="Button1" runat="server" Text="Submit" OnClick="Button1_Click" />
        <br /><br />

        <asp:Label ID="Label1" runat="server" ForeColor="Green"></asp:Label>

    </form>
</body>
</html>
using System;

public partial class Demo : System.Web.UI.Page
{
    protected void Page_Load(object sender, EventArgs e)
    {
        // Page load logic (optional)
    }

    protected void Button1_Click(object sender, EventArgs e)
    {
        Label1.Text = "Submit Button Clicked Successfully!";
    }
}
```

Output:

Hello John
City: Bangalore
Gender: Male

Skills: C#

Result:

A Web Application using Data controls has been created.

EX NO: 14 **Develop Web Applications using**
DATE: **Data Source Control**

Aim:

To Develop Web Applications using Data Source Control.

Code:

```
<%@ Page Language="C#" AutoEventWireup="true"
   CodeFile="DataSourceExample.aspx.cs"
   Inherits="DataSourceExample" %>

<!DOCTYPE html>
<html>
<head runat="server">
   <title>Data Source Control Example</title>
</head>
<body>
   <form id="form1" runat="server">

      <h2>Student Details</h2>

      <asp:GridView
         ID="GridView1"
         runat="server"
         AutoGenerateColumns="True"
         DataSourceID="SqlDataSource1">
      </asp:GridView>
```

```
 <asp:SqlDataSource
        ID="SqlDataSource1"
        runat="server"
        ConnectionString="<%$ ConnectionStrings:MyConnectionString %>"
        SelectCommand="SELECT * FROM Students">
      </asp:SqlDataSource>

   </form>
</body>
</html>
<configuration>
  <connectionStrings>
    <add
      name="MyConnectionString"
      connectionString="Data Source=.;Initial Catalog=StudentDB;Integrated Security=True"
      providerName="System.Data.SqlClient" />
  </connectionStrings>
</configuration>
```

Ouput:

Student Details

StudentID	Name	Age	Course
1	John	20	CS
2	Priya	21	IT
3	Ahmed	22	ECE

Result:

A Web Application using Data Source control has been created.